Biological positive controls for cross-cohort validation of prognostic gene expression, with a case study of EGFR in glioma

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Abstract

Prognostic gene associations from one cohort frequently fail to replicate in another, and it is hard to know in advance which will. We evaluate a simple safeguard—a *biological positive control*: a cross-cohort comparison is trustworthy only if both cohorts measure the gene faithfully, checkable by whether its expression tracks a known biological covariate (here tumor grade or *IDH* status) consistently across cohorts. Using The Cancer Genome Atlas (TCGA) as discovery and three Chinese Glioma Genome Atlas (CGGA) datasets for replication, concordance predicted replication across 6,512 TCGA-prognostic genes with **AUC** 0.82–0.93. Our central methodological result concerns how such a claim must be scored. Against a matched null—a held-out, same-population cohort of identical size, in which no cross-cohort artifact can exist—the score the field would naturally reach for, the anchor *product*, still reaches AUC 0.89–0.93, so its distance from chance carries no information; across five cohort pairs and two tumor types it beats its null in two settings and loses in three. The *difference* form $-|r_D - r_R|$ is far less confounded (null 0.53–0.61) and is positive in all three glioma pairs, including the one where the product form scores *below* its null, while a discovery effect-size control favors the null everywhere, as a negative control should. Quantifying uncertainty honestly is itself a finding here: between-split variance accounts for 77–84% of the total, so intervals from a single split understate it by 3–8×, and once both sources are combined only one of three glioma pairs excludes zero. We therefore report a consistent direction rather than three significant results. Under injected ground truth, detectability is strongly mode-dependent: compression of between-stratum signal—the signature EGFR shows in CGGA—is detected at AUC 0.73–0.85, permutation and additive noise at 0.63–0.86, and flooring essentially not at all (≤ 0.60). The gate is therefore worth running, in its difference form, calibrated against a split-half null rather than against chance, and not relied upon where saturation is the suspected failure. The framework is motivated by EGFR: prognostic in TCGA only until *IDH* is modeled (HR 1.31 $\rightarrow$ 1.13, $p = 0.16$), carrying no within-IDH-wildtype signal (HR 1.00, 0.89–1.13), and showing in all three CGGA datasets a 2–4× collapse in the share of its variance explained by grade and *IDH* that cohort composition does not account for.

1. Introduction

Diffuse gliomas vary widely in outcome, and their prognosis is defined primarily by molecular features—*IDH* mutation status and 1p/19q codeletion—rather than histology alone [5]. Against this backdrop, individual genes are frequently proposed as transcriptomic prognostic markers. EGFR is a natural candidate: a defining oncogene of the classical glioblastoma subtype and one of the most frequently altered genes in glioma [1, 6].

Any single-gene prognostic claim in glioma faces a specific threat: because molecular subtype governs both a tumor's transcriptional program and its outcome, a gene whose expression tracks subtype will appear prognostic even if it carries no information beyond subtype. A credible claim must therefore (i) survive adjustment for *IDH* status and molecular subtype, (ii) be robust to expression normalization, and (iii) replicate in an independent cohort—with the replication cohort itself verified to measure the gene faithfully.

Our primary contribution is **methodological**, and it has two parts. First, we evaluate a biological positive-control gate for cross-cohort prognostic validation at genome scale (6,512 genes, three replication cohorts, two platforms, two anchors), where a gene's expression–covariate concordance predicts whether its prognostic association replicates (AUC 0.82–0.93). Second, we show that **this evaluation design is not interpretable as stated**, and supply the calibration it needs. Across five cohort pairs and two tumor types, a same-population null reaches AUC 0.89–0.93 with the product-form score, so its distance from chance is uninformative; the omitted effect-size baseline captures much of the apparent signal; and the fix is a change of statistic, from the product $r_D \times r_R$ to the difference $-|r_D - r_R|$, which is far less confounded and is positive in all three glioma pairs, including the one the product form gets backwards—though once split variability is included only one of the three is conclusive. A controlled experiment with injected ground truth then establishes which failure modes the gate detects (compression, permutation, noise, contamination) and which it misses (flooring). We tested the scope of this calibration problem rather than asserting it: it is not universal across metrics or settings, and Table 5 reports where it does and does not bite. We delineate the gate's **scope** in a second tumor type (breast, TCGA-BRCA↔METABRIC), where concordant measurement leaves it little to flag. The framework is motivated and stress-tested by a **case study of EGFR**, for which we additionally provide (i) nested Cox models with discrimination and PH diagnostics showing the association vanishes after *IDH* adjustment, robust to normalization and to an independent raw-read reprocessing; (ii) a within-IDH-wildtype analysis of expression and copy number showing no within-subtype signal; and (iii) a decomposition separating what CGGA's cohort composition explains from what it does not.

2. Background and Related Work

TCGA molecularly characterized thousands of tumors with matched survival data [1, 2, 3]; its glioma studies—glioblastoma [1] and lower-grade glioma [2]—include curated molecular classifications (*IDH* status, 1p/19q codeletion, subtype) [2]. CGGA is an independent resource providing RNA-sequencing and clinical data, including *IDH* status and treatment, for a large Chinese glioma cohort [17]. Under the 2021 WHO classification adult diffuse gliomas are defined molecularly and *IDH* mutation is strongly favorable [5, 7], so "lower-grade glioma" spans prognostically distinct entities. EGFR amplification, mutation (EGFRvIII), and overexpression are recurrent in glioblastoma and mark the classical subtype [1, 6]; EGFR expression is characteristically higher in higher-grade and IDH-wildtype tumors—the fact we exploit as a positive control. Throughout we use Kaplan–Meier estimation with the log-rank test [9, 10], Cox regression [11], Harrell's C-index [16], PH testing via scaled Schoenfeld residuals [13], Benjamini–Hochberg FDR [12], and DESeq2 variance-stabilizing transformation [18].

Cross-study irreproducibility of transcriptomic prognostic signatures is well documented: Michiels et al. [24] found prognostic classifiers unstable under resampling across seven microarray studies, and Venet et al. [25] showed that in ER-positive breast cancer almost any gene set—including biologically irrelevant ones—is associated with outcome, because a dominant proliferation axis correlates with most genes. These motivate two distinctions our framework makes explicit: *reproducibility is not validity* (a reproducible association may still be confounded, as with EGFR and *IDH*), and *non-replication has multiple causes*—limited power, population differences, and measurement artifact. Four bodies of prior work sit closest to ours and delimit what is and is not new here. **sigQC** [26] standardizes quality control for gene *signatures* across datasets, checking expression, variability, autocorrelation, and scoring stability; it operates at the level of a gene set and does not anchor individual genes to a known biological covariate. **Removing Unwanted Variation** [27] uses *negative* control genes—genes assumed unassociated with the factor of interest—to estimate and subtract unwanted variation; we use the mirror-image construct, *positive* controls, and use them to gate rather than to correct. Third, the replication-prediction literature outside genomics has established that simple features of a discovery study, above all its **effect size**, predict replication well (AUC ≈ 0.77 in psychology and economics [28]); this is precisely the baseline our own genome-scale evaluation initially omitted, and §6.9 shows it recovers much of the apparent performance of a positive-control gate. Fourth, TCGA→CGGA validation is a heavily used design in glioma transcriptomics [29], which is what makes a per-comparison check on it practically consequential rather than merely methodological.

Against that background, what we ask is a per-gene, per-comparison question—does each cohort measure *this* gene faithfully, judged against known biology—and what we contribute is less the gate itself than the demonstration of how such a gate must be scored: against a same-population null rather than chance, using a difference rather than a product statistic, and validated against injected ground truth rather than observational agreement (§6.9).

3. Problem Statement

For TCGA-LGG, TCGA-GBM, and (for validation) CGGA, let z_{EGFR} be per-SD standardized EGFR expression. We ask whether z_{EGFR} is **associated** with overall survival within each cohort; whether any association is **independent** of age, grade, and *IDH*/subtype; whether EGFR improves **discrimination** (C-index) beyond established covariates; whether conclusions are **robust to normalization** (VST rather than TPM); and whether the TCGA conclusions **reproduce externally** in CGGA—given that CGGA must first be shown to measure EGFR faithfully enough to test them.

4. Methodology

4.1 Data acquisition and expression

TCGA RNA-sequencing and clinical data were retrieved from the Genomic Data Commons [3] with `TCGAbiolinks` [4] (harmonized data, GDC Data Release 45.0, 2025-12-04), using STAR-Counts quantifications [8]; molecular annotations came from the TCGA marker-paper fields [1, 2]. Three CGGA datasets were downloaded from the CGGA portal [17] with their clinical tables (all release 20200506): the mRNAseq_693 and mRNAseq_325 RSEM gene-level matrices and the mRNA-array_301 gene-level microarray matrix. TCGA expression used the TPM assay, $\log_2(\text{TPM}+1)$; the CGGA RSEM matrices used $\log_2(\text{RSEM}+1)$; the microarray matrix is distributed on a log-ratio scale and was used untransformed (§4.6). Expression was standardized to unit SD (z-scores) within each analysis sample, so the per-SD unit is defined by the patients entering a given model; the common-sample refit (§6.2) therefore uses a marginally different SD from Table 1. Genes were matched by HGNC symbol; in CGGA, EGFR was verified to be a unique feature. We retained primary solid tumors, one sample per patient. Survival models additionally require positive overall-survival (OS) time (time to death as event, else censored at last follow-up); the expression positive controls, which involve no survival model, use all such tumors with a recorded grade. Grade is handled asymmetrically: TCGA-LGG uses the recorded histologic grade (available for 454 of 511), while all TCGA-GBM patients are treated as WHO grade IV by study definition (281 of 282 carry a recorded grade). For CGGA, WHO grade II/III was the lower-grade analog and grade IV the glioblastoma analog.

4.2 Statistical analysis

Cox regression modeled $h(t \mid \mathbf{x}) = h_0(t)\exp(\boldsymbol{\beta}^\top\mathbf{x})$. For EGFR we fit nested models adding covariates stepwise: unadjusted; +age; +grade; +*IDH* status; +full IDH/1p19q subtype (TCGA). We report the EGFR HR per SD (HR >1 = worse survival), Harrell's C-index [16] with 95% CIs, and the PH test for the EGFR term [13]. So that attenuation across the nested models reflects the added covariate rather than the sample change induced by complete-case fitting, the models were also refit on a single common *IDH*-complete subset; EGFR's incremental value was assessed by a likelihood-ratio test and the change in C-index. Kaplan–Meier curves (median split) are descriptive [9, 10]. We treat *IDH* status as a confounder; because *IDH* mutation is an upstream driver and EGFR expression is partly downstream of the IDH-wildtype state, the adjustment could also remove a mediated path—either way, EGFR carrying no signal after conditioning on *IDH* means it adds no information beyond subtype. Ten EGFR/RTK–PI3K-axis genes (EGFR, IDH1, TP53, CDKN2A, NF1, PTEN, PDGFRA, PIK3R1, PIK3CA, RB1) were each entered per SD into age-adjusted and (in TCGA-LGG) age-plus-*IDH*-adjusted Cox models, with Benjamini–Hochberg FDR correction [12].

4.3 Normalization sensitivity and external validation

We recomputed the key TCGA EGFR models on DESeq2 VST-transformed STAR counts [18]. For CGGA we repeated the nested EGFR models (adding radiotherapy and chemotherapy status), and—before interpreting—assessed positive controls. The criterion was **specified a priori from established biology**: because EGFR expression is higher in higher-grade and IDH-wildtype glioma, a cohort that measures EGFR faithfully must reproduce a positive EGFR expression–grade correlation, as TCGA does. Clinical controls (IDH and grade must predict survival) verify the survival and annotation data independently of expression, and the between-cohort difference was tested formally (Fisher r-to-z). To distinguish a genuine CGGA property from an analyst error, we verified the expression–clinical join, reproduced the result independently, and repeated the controls in CGGA-325 and the array-301 microarray. To test pipeline robustness we re-obtained TCGA RNA-seq from recount3 [19]—an independent Monorail reprocessing of the raw reads—and repeated the nested models.

4.4 Within-subtype and copy-number analysis

To test whether EGFR is prognostic within IDH-wildtype glioma, we restricted TCGA (LGG+GBM) to IDH-wildtype tumors and fit Cox models for EGFR expression (per SD) and, separately, for gene-level EGFR copy number obtained from the GDC (ASCAT3 gene-level calls). High-level amplification was pre-specified as copy number ≥ 6, with sensitivity at CN ≥ 5 and ≥ 7 and a continuous analysis on $\log_2$ copy number (per SD). Grade throughout is the original histologic grade recorded by TCGA; because under the WHO 2021 scheme EGFR amplification is itself a grade-4 criterion [5], grade-adjusted amplification analyses carry a partial circularity that the pre-2021 grading mitigates. For the amplification analysis we report a power/precision analysis (detectable HR at 80% power via the Schoenfeld relation) alongside the point estimates.

4.5 Positive-control-gated screens

We first generalized the positive control to the 500 most variable genes shared by TCGA and CGGA, ranked by the standard deviation of $\log_2$ expression in TCGA. For each gene we recorded the age-adjusted Cox coefficient and the expression–grade Spearman correlation in each cohort; a gene was "QC-concordant" if its grade correlation had the same sign and (pre-specified) magnitude > 0.1 in both cohorts, and "QC-discordant" otherwise. Among genes prognostic in TCGA (FDR < 0.05), we computed cross-cohort replication (same sign, $p < 0.05$ in CGGA) and sign-flip rates, stratified by QC status, with binomial 95% CIs and a Fisher exact test on the QC $\times$ replication table. The criteria are deliberately asymmetric—FDR-controlled for discovery, nominal $p < 0.05$ for replication—the conventional, more permissive choice, which yields a high replication base rate; the QC contrast is a comparison *within* that base rate, not a claim about its level. We repeated the analysis in a second, independent cohort pair (TCGA→CGGA-325), varied the QC magnitude threshold ($|r| > 0.05$, 0.1, 0.2), and compared against a no-gating baseline. Throughout we use *QC-concordant* and *QC-discordant* for genes passing and failing this check; when the anchor is grade specifically we also call the latter *grade-discordant*, a deliberately neutral term, since the flag marks a candidate measurement artifact but does not by itself prove mis-measurement—a gene's grade relationship could differ biologically across populations.

To evaluate the framework at genome scale, we extended it to the genes shared by TCGA and all three CGGA cohorts, retaining those with TCGA expression SD > 0.5 and, for runtime, capping the universe at the 8,000 most variable; of these, 6,512 (81%) were prognostic in TCGA at FDR < 0.05 and form the evaluation set. Such a large majority qualifies because survival in pooled LGG+GBM is grade-dominated, so "prognostic" is a weak label here; the circularity control (§6.8) tests whether the gate survives removal of that shared grade signal. For each gene we scored positive-control concordance as the cross-cohort product of expression–grade Spearman correlations (grade anchor) and, as an alternative, of expression–*IDH* correlations (IDH anchor), and measured how well each score predicts replication by the area under the ROC curve (Mann–Whitney estimator). We benchmarked these biological anchors against two baselines: a *detectability* baseline (minimum cross-cohort mean expression) and a *precision* baseline (the standard error of the replication-cohort Cox coefficient), the latter to test whether the anchor merely proxies how precisely a gene's effect is estimated in the replication cohort. In addition to the detectability and precision baselines we benchmarked the anchor against the discovery-cohort effect size $|\beta_{\mathrm{TCGA}}|$, and assessed its incremental value by adding it to a logistic model for replication already containing effect size (likelihood-ratio test). To calibrate the AUC itself we constructed a matched null: within each cohort pair, one discovery set was evaluated against two replication cohorts of identical size—a disjoint, held-out subsample of the discovery cohort and the real replication cohort subsampled to match—over 8–10 replicates. This was repeated in five settings (TCGA→CGGA-693, -325, array-301; TCGA-BRCA↔METABRIC) and for five per-gene quality metrics: the anchor product $r_D r_R$, the anchor difference $-|r_D - r_R|$, expression-level concordance, dynamic-range concordance, and co-expression preservation over 100 hub genes, with $|\beta_D|$ as a negative control. Because the null arm replicates at a higher rate than the real arm, we also report a rate-matched variant in which the null arm's replication threshold is tightened until the two rates agree. To establish ground truth we split TCGA into halves and injected two mechanisms separately at known doses: a confounding-structure shift, by biasing assignment so grade and *IDH* are coupled more tightly in one half ($\delta = 0$–0.4), and measurement corruption in a random 20% of genes in one half only, scoring the gate against the genes actually corrupted (2,000 most-variable genes). Five corruption modes were used at three doses each: permutation of a fraction of values; additive Gaussian noise at k SD; compression of the between-stratum component toward the grand mean by a factor λ (which lowers between-stratum variance while preserving within-stratum variance, reproducing the signature observed for EGFR in CGGA); flooring, censoring values below the qth quantile; and contamination, mixing in a scale-matched unrelated gene at weight w. We also decomposed cross-cohort anchor discordance into a composition component and a residual, by transporting TCGA's within-(grade, *IDH*)-stratum means and variances onto each replication cohort's stratum weights; because grade is constant within a stratum this yields the implied correlation in closed form. Replication cohorts were CGGA mRNAseq_693, mRNAseq_325, and the mRNA-array_301 microarray; because these datasets are drawn from one resource, we quantified patient overlap between them and repeated the array-301 evaluation after removing shared patients (§6.8). To probe scope in a second tumor type, we repeated the framework in breast cancer using TCGA-BRCA (discovery) and METABRIC (replication); standardized expression and ER/subtype labels came from a common processed resource [20], overall survival from the respective cohort clinical files [21], and ER status and basal/non-basal subtype served as the biological anchors (in place of grade).

4.6 Reproducibility and multiplicity

The pipeline is deterministic except for the three resampling analyses (matched null, controlled experiment, bootstrap CIs), which are seeded (`set.seed(20260815)`). Data provenance is pinned: TCGA via the GDC **Data Release 45.0 (2025-12-04)**, and CGGA datasets mRNAseq_693, mRNAseq_325, and mRNA-array_301 (all release 20200506). CGGA data were used under the CGGA data-use terms, which permit academic reuse with citation [17]. All positive-control and amplification thresholds ($|r| > 0.1$; CN ≥ 6) were pre-specified from biology, not tuned to outcomes. Expression matrices were transformed on their native scales: $\log_2(x + 1)$ for the TCGA TPM and CGGA RSEM matrices, and no transformation for the CGGA mRNA-array_301 matrix, which is already log-ratio scaled and contains negative values (applying $\log_2(x + 1)$ to it discards every value ≤ -1). Analysis scripts (steps 00–32; steps 21–22 are independent verification suites that reconstruct every reported sample size and refit every headline model, steps 23–31 are the calibration, controlled-experiment, transport, baseline and breast-verification analyses added in revision, and step 32 audits every figure transcribed into §6.4 and §6.9 and Tables 5–6 against its source output, 120 checks). Figures rendered by steps 25, 28, 30 and 31 are not reproduced here for space; they are in the repository and duplicate the content of Tables 5 and 6, their console outputs, and full session information are available at https://github.com/edanmn/TCGA-EGFR-Glioma; the workflow is summarized in Figure S1 and software versions in the Supplementary Methods. The study reports one prespecified primary hypothesis (EGFR → overall survival, adjusted) evaluated across cohorts and robustness checks; these primary p-values are nominal and should be read as a small prespecified family. The pathway and systematic screens are explicitly exploratory and FDR-controlled within each analysis; the single nominally significant secondary observation (the inverse EGFR trend in IDH-wildtype glioblastoma, HR 0.87, $p = 0.034$; Table 3) is not corrected and is not interpreted as confirmatory.

5. Experimental Setup

After quality control, TCGA-LGG comprised 511 patients (125 deaths) and TCGA-GBM 282 (227 deaths); molecular composition differs sharply (Table S9). CGGA-693 contains 422 primary tumors with a recorded WHO grade—the set used for the expression positive controls (Table 2)—of which 404 also have evaluable overall survival and enter the survival models: 271 lower-grade (WHO II/III, 99 deaths) and 133 glioblastoma (WHO IV, 110 deaths). Analyses used R 4.6.0 [15] with `TCGAbiolinks` 2.40.0 [4], `SummarizedExperiment` 1.42.0, `DESeq2` [18], `survival` 3.8-6 [13], and `survminer` 0.5.2 [14].

6. Results

6.1 Unadjusted association in TCGA (descriptive)

Under a median split, high EGFR expression was associated with shorter survival in TCGA-LGG (median 2,988 vs. 2,282 days; log-rank $p = 0.040$) and pooled ($p = 4.85 \times 10^{-3}$), but not TCGA-GBM ($p = 0.99$) (Table S1; Figures S2–S4).

6.2 In TCGA, EGFR is not independent of IDH status

The nested Cox models (Table 1) show that in TCGA-LGG the EGFR association—strong when unadjusted (HR = 1.59 per SD) and after age and grade (HR = 1.31; $p = 8.2 \times 10^{-3}$)—was **abolished after adjustment for *IDH* status** (HR = 1.13; $p = 0.16$) and molecular subtype (HR = 1.14; $p = 0.10$). EGFR alone discriminated poorly (C-index 0.59, 95% CI 0.51–0.67) versus 0.83 (0.79–0.87) for age, grade, and *IDH*.

To confirm that the attenuation reflects *IDH* rather than the sample change from complete-case fitting, we refit the models on the identical *IDH*-complete subset ($n = 452$, 105 events): EGFR remained significant after age and grade (HR = 1.308; 1.072–1.595) and lost significance only when *IDH* was added (HR = 1.13; 0.95–1.34; $p = 0.16$). A likelihood-ratio test confirmed that EGFR adds nothing beyond age, grade, and *IDH* ($\chi^2 = 1.98$, df = 1, $p = 0.16$), and it did not improve discrimination: C-index 0.834 (0.79–0.87) without EGFR versus 0.830 (0.79–0.87) with it.

The EGFR term violates proportional hazards here (PH $p = 0.004$), so we checked the null does not depend on a constant-hazard summary. A time-varying effect gave a nominal interaction with log time ($p = 0.045$) and no significant main effect ($p = 0.080$); splitting follow-up at three years gave HR = 0.96 (0.81–1.14, $p = 0.65$) before and 1.54 (0.95–2.51, $p = 0.079$) after. EGFR is thus not prognostic in either period, but the effect is not constant and a late-emerging association cannot be excluded.

In TCGA-GBM, EGFR was not associated when unadjusted; a weak inverse association in the age- and *IDH*-adjusted model (HR = 0.87; $p = 0.030$; $n = 265/214$—a whole-cohort model distinct from the IDH-wildtype-restricted analysis of §6.6) was of borderline significance, in the opposite direction, and not robust across model choices, so we do not interpret it as a real effect. The pooled association (HR = 1.35; $p = 1.98 \times 10^{-7}$) vanished after age adjustment, demonstrating confounding.

6.3 Subtype drives the association; pathway screen and normalization checks

In TCGA-LGG survival separates strongly by molecular subtype (Figure S5) and EGFR expression is elevated in the IDH-wildtype group, so its unadjusted association reflects subtype. In the age-adjusted pathway screen (Table S2) six genes were significant; after adding *IDH*, EGFR, PIK3R1, and PTEN lost significance while TP53, RB1, IDH1, CDKN2A, and PIK3CA remained, consistent with subtype-driven signal (IDH1 *expression* is distinct from the *IDH* mutation). No gene was significant in TCGA-GBM. Recomputing on DESeq2 VST counts reproduced the conclusion (Table S3): the lower-grade EGFR HR fell from 1.50 to 1.11 ($p = 0.21$) after age, grade, and *IDH*, so the pattern is not a TPM artifact.

6.4 External validation is inconclusive: CGGA fails EGFR positive controls

A naive CGGA analysis appeared to show the opposite of TCGA: EGFR was null unadjusted (HR = 1.17; $p = 0.16$) but *significant after* adjusting for age, grade, and *IDH* (HR = 1.44; $p = 4.6 \times 10^{-4}$), and after treatment (HR = 1.37; $p = 6.8 \times 10^{-3}$) (Table S8). Before interpreting this as a genuine discordance, we tested positive controls.

The CGGA **clinical** controls are strong and correct: IDH-mutant status is highly protective (HR = 0.23) and grade predicts survival (grade IV vs. II HR = 8.0), confirming the survival, grade, and *IDH* annotations and the sample–clinical join are sound. But the **EGFR-expression** controls fail. EGFR should rise with grade and be higher in IDH-wildtype tumors, as in TCGA; in CGGA it does neither (Table 2; Figure 1). EGFR expression is essentially flat across WHO grade (Spearman $r = -0.03$, vs. +0.18 in TCGA) and is *higher* in IDH-mutant tumors (mean $\log_2$ 4.17 vs. 3.63; the reverse of TCGA-LGG's 6.43 vs. 7.44, or 6.43 vs. 7.52 pooled). Expression–grade correlations are systematically attenuated across the whole gene panel in CGGA relative to TCGA (Table S4), indicating a general weakening of expression–phenotype signal in this matrix rather than an EGFR-specific mislabeling.

Composition is a real part of this, but not all of it. CGGA's grade strata are less *IDH*-differentiated than TCGA's (grade IV: 18%, 13%, 13% IDH-mutant across the three CGGA datasets vs. 8% in TCGA; grade II: 76%, 89%, 71% vs. 91%), so some of the flat marginal grade correlation is a difference in patient mix rather than in measurement. Holding *IDH* fixed, the EGFR–grade slope in CGGA-693 (+0.124 per SD, $p = 0.08$) is statistically indistinguishable from TCGA's (+0.115, $p = 0.065$), and within IDH-wildtype tumors the EGFR–grade correlation is *positive* in all three CGGA datasets (+0.163, +0.065, +0.079)—the negative marginal values are partly an artifact of *IDH* mixing. Transporting TCGA's within-stratum structure onto each CGGA cohort's (grade, *IDH*) composition confirms this is a minority effect genome-wide: composition accounts for only 8–25% of the mean cross-cohort anchor discordance, and the transported prediction does not track CGGA's actual anchor correlations better than TCGA's raw ones do ($r = 0.82$–0.87 vs. 0.81–0.88).

What composition cannot explain. Decomposing EGFR's variance across the six (grade, *IDH*) strata separates the two accounts cleanly. The share of EGFR's variance explained by those strata collapses from 0.106 in TCGA to 0.045, 0.030, and 0.024 in CGGA-693, -325, and array-301, while the *within*-stratum SD is essentially unchanged (1.88 vs. 1.84, 1.84, 1.84). EGFR varies as much as ever in CGGA; its variation is simply far less related to the biology. Reweighting cannot produce this—the transport calculation, which holds TCGA's conditional structure fixed and applies CGGA's stratum weights, predicts a *higher* correlation than TCGA's own (+0.24 vs. +0.22), against an observed +0.035, leaving a residual of −0.21 in all three datasets. Note also that additive noise would inflate the within-stratum SD, which it does not; the pattern is a compression of biological signal, consistent with quantification of a different feature rather than with a noisier assay. The one composition-immune anomaly that reaches significance is the reversed *IDH* relationship in CGGA-693 with grade held fixed (+0.389, $p = 8.8 \times 10^{-4}$, vs. TCGA's −0.402, $p = 8.7 \times 10^{-5}$); this does *not* reproduce in CGGA-325 (−0.003, $p = 0.99$) or array-301 (+0.158, $p = 0.29$), so the claim of a reproducible failure holds for the attenuation of biological signal but not for the sign reversal.

Two further checks establish this is a genuine property of CGGA's EGFR quantification rather than a processing error. First, the expression–clinical join is complete—all 693 samples match a clinical record, of which 422 are primary tumors with a recorded grade—and the flat EGFR–grade relationship was reproduced by an independent computation. Second, the failure **reproduced in two further CGGA datasets spanning a different platform**: a second RNA-seq batch (mRNAseq_325, $n = 229$) and a microarray (mRNA-array_301, $n = 264$). In all three datasets

Table 1: Nested Cox models for EGFR expression in TCGA (HR per 1 SD), with Harrell's C-index and the PH test p-value for the EGFR term. Grade is omitted for glioblastoma.

Cohort	Model	HR (95% CI)	p	C-index	PH p (EGFR)	n / events
TCGA-LGG	Unadjusted	1.59 (1.30–1.94)	6.85×10^{-6}	0.592	0.98	511 / 125
TCGA-LGG	+ age	1.37 (1.14–1.66)	1.06×10^{-3}	0.744	0.88	511 / 125
TCGA-LGG	+ age, grade	1.31 (1.07–1.59)	8.22×10^{-3}	0.786	0.23	454 / 106
TCGA-LGG	+ age, grade, IDH	1.13 (0.95–1.33)	0.164	0.830	0.004	452 / 105
TCGA-LGG	+ age, subtype	1.14 (0.98–1.34)	0.096	0.820	0.012	508 / 123
TCGA-GBM	Unadjusted	0.962 (0.845–1.09)	0.552	0.523	0.19	282 / 227
TCGA-GBM	+ age, IDH	0.866 (0.761–0.986)	0.030	0.639	0.40	265 / 214
Pooled	Unadjusted	1.35 (1.20–1.51)	1.98×10^{-7}	0.553	0.03	793 / 352
Pooled	+ age, grade, IDH	0.921 (0.840–1.01)	0.078	0.837	0.56	717 / 319

the EGFR–grade correlation differed significantly from TCGA (Fisher r-to-z $p < 10^{-3}$; Table 2), while clinical controls passed in every one (IDH-mutant HR 0.22–0.32; high-grade HR 7–9). The criterion was specified a priori from biology, not after seeing the disagreement, and the failure holds across two RNA-seq batches and a microarray—so it is a property of EGFR measurement in this resource, not one pipeline's quantification.

The apparent "IDH-independent EGFR effect" therefore cannot be read as genuine, and CGGA does not provide a valid external test for this gene; a naive reading would have been a spurious positive that only the positive controls exposed. Identifying the mechanism would require raw-read reprocessing, but consistency across two independent batches points to a systematic processing choice rather than random noise.

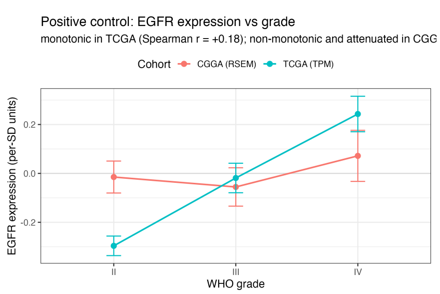

Figure 1. Positive control: standardized EGFR expression by WHO grade. EGFR rises monotonically with grade in TCGA (Spearman $r = +0.18$; canonical biology). In CGGA the relationship is *non-monotonic* rather than level—it falls from grade II to III ($-0.015 \rightarrow -0.055$) and rises from III to IV ($\rightarrow +0.072$), with overlapping standard errors—so the near-zero rank correlation ($r = -0.03$) reflects the absence of a consistent gradient, not a flat line. Points are means, error bars $\pm$ standard error. The between-cohort difference is significant (Fisher r-to-z $p = 4.5 \times 10^{-4}$). TCGA $n = 739$ (grade II: 216, III: 241, IV: 282); CGGA-693 $n = 422$ (II: 138, III: 144, IV: 140).

6.5 The TCGA result is robust to the processing pipeline

To test whether the finding depends on the GDC pipeline, we re-obtained TCGA from recount3 [19], an independent Monorail reprocessing of the raw reads. EGFR again passed the positive controls—expression rose with grade and was higher in IDH-wildtype tumors (8.92 vs. 7.99 in lower-grade; 9.20 vs. 7.63 in glioblastoma)—and the confounding result replicated: the lower-grade EGFR HR fell from 1.54 (unadjusted) to 1.27 (age+grade) to 1.11 (0.94–1.31; $p = 0.22$) after *IDH* (Table S5), matching the GDC-pipeline (Table 1) and VST (Table S3) estimates. The result is thus not a processing artifact, and a faithfully reprocessed cohort recovers the EGFR biology CGGA lacks. CGGA raw reads are controlled-access, so the equivalent reprocessing there—the only route to new-patient external validation—remains the key open step.

6.6 EGFR is not prognostic *within* IDH-wildtype glioma

A stronger test than confounder adjustment is whether EGFR carries any prognostic signal *within* the aggressive IDH-wildtype subtype, where subtype confounding no longer applies. Among IDH-wildtype gliomas (TCGA LGG+GBM; $n = 338$, 253 deaths), EGFR expression was not prognostic (Table 3): unadjusted HR = 1.00 (0.89–1.13), age-adjusted 0.93 ($p = 0.22$), and +grade 0.90 ($p = 0.10$), with a weak inverse—not adverse—trend in IDH-wildtype glioblastoma (HR = 0.87, $p = 0.034$, not corrected for multiplicity). This is a well-powered null: with 253 events, 80% power extends to HR ≥ 1.19 per SD, and the observed interval excludes HR > 1.13. The PH assumption held for the EGFR term ($p = 0.084$). Grade here is the *original histologic* grade recorded by TCGA (pre-2021), not the molecular WHO 2021 grade.

Gene-level copy number for the same tumors tells the same story. Of the primary tumors with both copy-

Table 2: EGFR positive control across TCGA and three CGGA datasets spanning two platforms, all computed by one method: Spearman correlation of EGFR expression with WHO grade among primary tumors with a recorded grade; Fisher r-to-z against the TCGA reference; univariable IDH-mutant HR; and grade IV vs. grade II HR (grade II and IV patients only). The correlation is positive in TCGA but null/negative in every CGGA dataset, whereas clinical controls are strong and correct throughout. The expression-level failure is reproducible across batches and platforms; the clinical data are sound.

Cohort	r(EGFR, grade)	n	p vs. TCGA	IDH-mutant HR	Grade IV vs. II HR
TCGA (reference)	+0.179	739	—	0.11	14.8
CGGA-693 (RNA-seq)	−0.033	422	4.5×10^{-4}	0.23	8.0
CGGA-325 (RNA-seq)	−0.097	229	2.5×10^{-4}	0.22	9.3
CGGA-301 (microarray)	−0.086	264	2.0×10^{-4}	0.32	7.3

number calls and *IDH* status, high-level EGFR amplification was strongly enriched in IDH-wildtype disease (55% of 167, vs. 2.3% of 222 IDH-mutant), confirming the copy-number data are faithful—a positive control the CGGA expression data failed. The 166 IDH-wildtype tumors that also have evaluable survival form the Cox models below. Within IDH-wildtype glioma we found *no evidence* that amplification predicts survival: the continuous effect on $\log_2$ copy number was null (HR = 1.00 per SD, 0.84–1.20, $p = 0.98$, age-adjusted), and binary high-level amplification was null across thresholds (CN $\geq$ 5/6/7: HR 1.09/1.00/0.92, all $p > 0.6$; Table 3). The binary tests are underpowered—with 127 events and 55% amplified, 80% power (two-sided $\alpha = 0.05$) extends only to HR $\geq$ 1.65—so we exclude a large but not a modest amplification effect. The continuous analysis is better powered (80% power for HR $\geq$ 1.28 per SD) and its interval excludes that value; and the expression analysis, which captures amplification's downstream impact, shows a precise null. EGFR therefore carries no detectable within-subtype prognostic signal.

6.7 A positive-control-gated framework for cross-cohort validation

The CGGA episode motivates a general safeguard: before trusting a cross-cohort comparison, verify both cohorts measure the gene faithfully. We formalized this on the 500 most-variable genes shared by TCGA and CGGA, with the expression–grade correlation in each cohort as a *pre-specified* positive control. Of the genes prognostic in TCGA (FDR < 0.05), replication in CGGA-693 was 92.8% (95% CI 89.9–95.0) among QC-concordant genes versus 24.3% (11.8–41.2) among QC-discordant genes (Fisher exact $p = 4.9 \times 10^{-21}$, OR = 39.2); both sign-flips observed in the panel fell among QC-discordant genes, though with only two such events this is suggestive rather than decisive (Table S6; Figure 2). The contrast reproduced in a second cohort pair with no shared patients (TCGA→CGGA-325: 95.4% vs. 19.2%; Fisher $p = 1.1 \times 10^{-20}$) and exceeded the no-gating baseline (87.3%, 84.0–90.2); 88% of the 500-gene panel passed QC. The contrast persisted across QC thresholds but narrowed as the threshold became more stringent, since a stricter bar moves well-measured genes into the failing set (QC-concordant vs. discordant replication: 91.0% vs. 5.0% at $|r| > 0.05$; 92.8% vs. 24.3% at 0.1; 96.2% vs. 53.6% at 0.2). For EGFR the artifact interpretation is independently supported by its canonical grade/IDH biology (§6.4). Gating thus concentrates non-replicating associations into a small, flaggable minority.

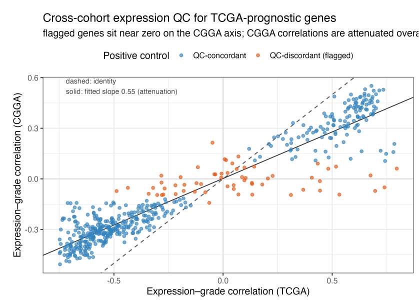

Figure 2. Cross-cohort expression QC for TCGA-prognostic genes. Each point is a gene; axes are its expression–grade correlation in TCGA (x) and CGGA (y). The dashed line is the identity; the solid line is the fitted slope of 0.55, showing that CGGA correlations are attenuated relative to TCGA's across the whole panel. Flagged genes (orange), including EGFR, are those sitting near zero on the CGGA axis, and are where cross-cohort prognostic replication breaks down. The flag marks discordance, not a demonstrated cause (§6.4).

6.8 The positive control predicts replication genome-wide

To test whether the positive control is a general safeguard rather than an EGFR-specific observation, we scaled it to **6,512 genes prognostic in TCGA** (FDR < 0.05, age-adjusted) and asked, for each, whether its association replicated in three CGGA cohorts (Table

Table 3: EGFR is not a within-subtype prognostic factor in IDH-wildtype glioma (TCGA). Expression (per 1 SD) shows a well-powered null; copy-number amplification shows no evidence of an effect (continuous, reasonably powered) and null across binary thresholds, though binary tests are underpowered (80% power only for HR ≥ 1.65). Grade is the pre-2021 histologic grade.

Marker	Model	HR (95% CI)	p	n / events
Expression	Unadjusted	1.00 (0.89–1.13)	0.99	338 / 253
Expression	+ age	0.93 (0.82–1.05)	0.22	338 / 253
Expression	+ age, grade	0.90 (0.80–1.02)	0.10	330 / 246
Expression	IDH-wt GBM, + age	0.87 (0.76–0.99)	0.034	244 / 203
Amplification	$\log_2$ copy number (per SD), +age	1.00 (0.84–1.20)	0.98	166 / 127
Amplification	CN ≥ 5 vs. not, +age	1.09 (0.76–1.57)	0.63	166 / 127
Amplification	CN ≥ 6 vs. not, +age	1.00 (0.70–1.42)	0.98	166 / 127
Amplification	CN ≥ 7 vs. not, +age	0.92 (0.65–1.32)	0.66	166 / 127

4; Figure 3). The grade anchor predicted replication with **AUC** 0.82 **(CGGA-693),** 0.93 **(CGGA-325), and** 0.92 **(microarray-301)**; the *IDH* anchor gave AUC 0.77–0.84. A simple grade-concordance flag identified non-replicating associations with 70% precision and 60% recall in CGGA-693, and 32% of the 6,512 prognostic genes were flagged.

Both of the baselines we originally chose are uninformative by comparison. Detectability (mean expression) ran at or near chance (AUC 0.36–0.53), so highly expressed genes are not more replicable, and *estimation precision* in the replication cohort—the standard error of its Cox coefficient—also failed to predict replication (AUC 0.44–0.59).

These two baselines do not, however, license the conclusion we initially drew from them. Both measure *noise*, and the objection they were meant to answer—that the anchor is a proxy for something other than measurement fidelity—is really about *signal*: a gene with a strong anchor relationship has a large survival effect, and large effects replicate. The correct competitor is therefore the discovery-cohort effect size, not the replication-cohort standard error. Section 6.9 supplies it, together with a calibrated null, and shows that both substantially qualify the reading of this table: the effect-size baseline reaches AUC 0.74–0.78, and a same-population null cohort produces an anchor AUC of 0.89–0.93 under the product-form score used here. The anchor retains incremental value over effect size, but the appropriate comparison for the numbers below is that null, not chance.

The three datasets come from one resource and are not fully independent: CGGA-693 and CGGA-325 share no patients, but of the 264 analysed array-301 patients, 61 (23%) also appear in CGGA-325 and 6 in CGGA-693. Removing all 67 leaves 197 patients and barely moves the discrimination—grade-anchor AUC 0.91 (0.91–0.92) versus 0.92 (0.92–0.93)—though the replication rate falls from 0.68 to 0.58, as expected from the smaller sample. The cross-platform finding therefore does not depend on shared patients, and EGFR's positive-control failure persists in the de-duplicated set ($r = -0.084$; Fisher $p = 6.6\times10^{-4}$). EGFR falls in the flagged, grade-discordant minority. Section 6.4 shows that its flag is driven by a genuine collapse in biologically-explained variance rather than by cohort composition, but the flag itself does not distinguish the two causes.

Controlling for circularity. Because survival in pooled glioma is grade-driven, the prognostic set is enriched for grade-correlated genes, and the grade-anchor QC could predict replication partly *because* both prognostic status and replication share the grade signal it measures. We tested this directly (Table S7). Restricting to associations that remain prognostic *after adjustment for grade and IDH*—i.e., subtype-independent signal—the QC still predicted replication above chance, though this set is small and the interval correspondingly wide (183 genes; AUC = 0.65, 95% CI 0.55–0.74). It also predicted replication for genes with low grade-correlation ($|r| < 0.2$; AUC = 0.75, 0.72–0.78). The AUC is higher for strongly grade-correlated genes (0.84) than for low-grade-correlation genes (0.75), so part of the headline value does reflect shared grade signal; but the effect persists for subtype-independent associations, indicating the positive control captures measurement fidelity *beyond* the anchor covariate itself rather than acting tautologically.

Figure 3. Anchor scores for predicting replication of TCGA-prognostic genes in each CGGA cohort, by metric. The grade and IDH anchors exceed both the detectability and estimation-precision baselines and the

Table 4: Genome-scale evaluation: a gene's biological positive-control concordance ranks whether its TCGA prognostic association replicates in a separate cohort (AUC). **These AUCs must be read against the same-population null of** 0.89–0.93 **established in §6.9, not against chance**, and against a discovery effect-size baseline of 0.74–0.78 that is not shown here; the detectability and precision columns measure noise rather than signal and are not the informative comparison. The three replication datasets come from one resource and are not fully independent (§6.8 quantifies and controls for patient overlap). $n = 6,512$ TCGA-prognostic genes (FDR < 0.05); AUCs are given to three decimals here and rounded to two in the text. Bootstrap 95% CIs are tight given $n \approx 6,500$ (CGGA-693 grade anchor: 0.815, 0.81–0.83); Table S7 gives the circularity control, Table 5 the matched-null calibration, and Table 6 the controlled experiment.

Replication cohort	Replication rate	AUC (grade)	AUC (IDH)	AUC (detectability)	AUC (precision)
CGGA-693 (RNA-seq)	0.622	0.815	0.771	0.358	0.591
CGGA-325 (RNA-seq)	0.701	0.931	0.839	0.366	0.439
CGGA-301 (microarray)	0.684	0.924	0.842	0.525	0.505

dotted chance line at 0.5—but chance is not the appropriate null. Red segments mark the same-population matched null for the product-form anchor (§6.9): 0.900, 0.893, 0.793. The grade anchor clears its null in CGGA-325 and array-301 and falls short of it in CGGA-693, which is the comparison the figure is drawn to make.

6.9 Calibration: what the gate detects, and how its metric must be scored

An AUC is interpretable only against the right null. Table 4 compares its anchors to chance (0.5), which presumes that a score can predict replication only by detecting something wrong. That presumption is false, and correcting it changes both how Table 4 should be read and which form of the statistic should be used.

A matched null, in five settings. We fixed one discovery set and gave it two replication cohorts of *identical size*: a held-out, disjoint subsample of the discovery cohort itself—same population, same platform, same processing, so no cross-cohort artifact can exist—and the real replication cohort subsampled to match. Repeating this across five cohort pairs spanning two tumor types and three platforms (8 replicates each; Table 5) shows that the score the paper uses, the anchor *product* $r_D \times r_R$, is **severely confounded**: against a null where nothing is wrong it still reaches AUC 0.79–0.90 in glioma and 0.72 in the one evaluable breast direction. Because a matched null requires splitting the discovery cohort, Table 5 necessarily runs at lower discovery power and on a smaller gene universe than Table 4 (2,000 vs. 6,512 genes; discovery $n = 368$–1,066 vs. 793), so its absolute AUCs are not interchangeable with Table 4's. The comparison that matters is *within* each row, where the two arms share a discovery set, a gene universe, and a replication-cohort size by construction. An AUC in the range Table 4 reports therefore does not by itself indicate that anything was detected.

It does not, however, follow that nothing was detected. The comparison goes both ways across settings: for CGGA-693 the null *exceeds* the real cohort (-0.075; CI -0.093 to -0.058), but for CGGA-325 and array-301 the real cohort exceeds the null ($+0.044$ and $+0.132$; both intervals excluding zero). The one evaluable breast direction favors the null (-0.150). Tightening the null arm's replication threshold until its replication rate matches the real arm's leaves these conclusions unchanged (e.g. CGGA-693 null $0.931 \rightarrow 0.921$), so the effect is not an artifact of differing base rates.

Robust to thresholds, sensitive to the anchor. The chain above rests on choices we made once: BH < 0.05 for "prognostic", nominal $p < 0.05$ for "replicated", and grade as the anchor. Varying the first two over $\{0.01, 0.05, 0.10\}$ each gives 9 threshold combinations per cell; across the three glioma settings and both metrics, **all 54 grade-anchor cells reproduce the sign reported here**, so the conclusions are not an artifact of our cutoffs. The anchor is a different matter: substituting *IDH* for grade, only 39 of 54 cells agree, and CGGA-325's product form inverts in all 9 (0% agreement). The calibration in this section is therefore established for the grade anchor specifically and should not be assumed to transfer to another anchor without re-running it—a caveat that applies to Table 4's *IDH* column as well, which was never calibrated.

The intervals were the wrong ones, and the corrected version is much wider. Two sources of variability sit outside a single-split gene bootstrap, and both matter. Co-expression first: transcriptomes come in correlated blocks, so resampling genes independently treats correlated observations as independent. Re-running the comparison as a *cluster* bootstrap—partitioning each evaluation set into 200 co-expression clusters (median 26–29 genes) and resampling whole clusters—widens the intervals by a median factor of 2.2 (range 1.6–2.7); on the identical split, CGGA-693's difference form moves from $+0.138$ ($+0.100$, $+0.176$) to ($+0.076$, $+0.200$). Split variability second, and it dominates. Repeating the whole procedure over $K = 6$ independent splits per setting and combining by the standard rule for repeated sampling ($\mathrm{Var}_{\mathrm{total}} = \overline{\mathrm{Var}}_{\mathrm{within}} + (1+1/K)\,\mathrm{Var}_{\mathrm{between}}$) attributes 77–84% **of total variance to between-split variation**: which patients land in which half matters far more than which genes are sampled. Combined intervals are 3.1–8.2× wider than the gene-bootstrap-only ones (median 6.5×), and difference-form point estimates range from +0.030 to +0.283 across splits.

Under that accounting the conclusions weaken to the following. The difference form stays positive in all three glioma pairs (+0.063, +0.198, +0.170, and positive in every one of the 18 split-level estimates), but only CGGA-325 excludes zero (+0.091 to +0.306); array-301 sits on the boundary (−0.008 to +0.348) and CGGA-693 spans it (−0.036 to +0.162). The product form is conclusive only in CGGA-693, where it is inverted (−0.183 to −0.011). **We therefore claim a consistent direction, not three significant results:** the difference form is positive wherever we measure it, and one of three cohort pairs reaches conventional significance once split variability is included. Table 5's single-split intervals substantially overstate the precision available from cohorts of this size, and are retained only because they are the per-metric comparison; the combined intervals here are the ones to quote.

We report intervals rather than tests across replicates for a reason worth stating, because it is an easy error to make. Our first analysis compared the arms with a t-test over 8 resampling replicates. Those replicates are resamples of one fixed cohort, so their spread is Monte-Carlo error of the estimate, shrinking as $1/\sqrt{\text{replicates}}$: re-running that comparison with 3, 4, …, 8 replicates moves p from 0.32 to 0.010 while the effect moves only from +0.030 to +0.037. Such a p-value measures how long we chose to run the simulation. The AUC's sampling unit is the gene, so all intervals reported here are paired bootstraps over genes.

The confound is in the functional form, and it is fixable. The product $r_D \times r_R$ is large when *both* correlations are large, so it rewards anchor magnitude—which tracks effect size—as much as cross-cohort agreement. The natural alternative, $-|r_D - r_R|$, scores only disagreement. Its behavior is markedly better: its null AUC is 0.53–0.61 in glioma rather than 0.79–0.90, i.e. it is far less confounded, and on a single split its point estimate exceeds the null in **all three** glioma pairs (+0.138, +0.185, +0.148). Notably it recovers the one setting the product form gets wrong: in CGGA-693 the product form scores *below* its null (−0.075) while the difference form scores above it (+0.138). We recommend the difference form over the product form on that basis, subject to the variance accounting above, which shows that only one of the three differences is statistically conclusive once split variability is included. Restricting to the 2,000 most variable genes leaves every other glioma conclusion unchanged but renders CGGA-693's difference form interval-spanning (+0.051; −0.019 to +0.121), so that particular cohort's result depends on having the full universe. This is the practical correction that follows from the calibration: not to abandon the gate, but to score it with a statistic that is not dominated by signal strength.

The anchor is nevertheless not redundant with effect size: adding it to a logistic model for replication that already contains $|\beta_D|$ raises the model AUC by +0.035 to +0.199 across the three glioma cohorts and both replication criteria (all likelihood-ratio $p < 10^{-37}$; §4.5).

A negative control behaves as it should. The discovery effect size $|\beta_D|$, which cannot by construction carry any cross-cohort information, scores *higher* against the null than against the real cohort in all four evaluable settings (−0.035 to −0.077), every interval excluding zero. That it separates in the expected direction, while the difference-form anchor separates in the opposite one in all three glioma pairs, is what licenses reading the latter as genuine cross-cohort signal rather than as more of the same confounding.

Controlled ground truth: which failure modes the gate catches. To establish what the gate can detect when the answer is known, we split TCGA into halves and corrupted a random 20% of genes in one half only, using five mechanistically distinct failure modes at three doses each (Table 6). Detectability differs sharply by mode. *Compression* of the between-stratum signal is the best detected (AUC 0.85, 0.73, 0.64 at $\lambda = 0.25$, 0.50, 0.75). It is also the mode that matches EGFR's behavior in CGGA, though identifying it requires two statistics rather than one. On variance share alone the match is not unique: at $\lambda = 0.50$ compression leaves 0.136 against the uncorrupted baseline of 0.339—a ratio of 0.40, inside EGFR's observed 0.23–0.42 band—but permutation at 50% (0.28) and noise at $k = 2$ (0.23) also fall in that band. What separates them is the *within*-stratum variance, which compression preserves by construction while permutation and added noise both inflate it; EGFR's within-stratum SD in CGGA is unchanged from TCGA's (1.84 vs. 1.88, §6.4), which is consistent with compression and not with the other two. On that two-statistic match the gate would have flagged an EGFR-like failure at AUC 0.730—an inference no observational analysis here could supply, but one that rests on a signature match rather than on a known cause. Permutation (0.63–0.86), additive noise (0.57–0.78), and paralog contamination (0.54–0.73) are also detected in a dose-dependent way. **Flooring—censoring low values, as a detection limit or sat-**

urating assay would—is a blind spot: AUC 0.50, 0.53, 0.60 across doses, barely above chance even where corrupted genes' replication falls to 0.82 against 0.95 for clean genes. A gate of this kind should not be relied on where saturation is the suspected failure.

Composition shift is not what drives the AUC. Biasing the split so that grade and *IDH* couple more tightly in one half than the other—reproducing under control the difference measured between TCGA and CGGA (§6.4)—does not raise the anchor's AUC. It lowers it, from 0.943 at no shift to 0.609 at the largest, because replication itself collapses (0.957 → 0.450) as the coupling difference grows from 0.045 to 0.796.

6.10 Scope: a well-matched second tumor type

To test whether the framework is a *universal* replication predictor or a targeted detector of measurement-artifact discordance, we applied it to a second, unrelated tumor type: breast cancer, using TCGA-BRCA [23] (discovery, RNA-seq) and METABRIC [22] (replication, microarray), with ER status and basal/non-basal subtype as biological anchors, expression from a common processed resource [20] and survival from the cohort clinical files [21]. Here the positive control was at most weakly predictive. In the TCGA-BRCA→METABRIC direction the ER-anchor AUC was 0.60 (95% CI 0.44–0.75), an interval that includes chance and rests on only 59 prognostic genes; in the better-powered reverse direction (2,568 genes) it was 0.56 (0.52–0.59), which *excludes* chance but is far below the 0.82–0.93 seen in glioma. The basal anchor reached at most 0.54. In the better-powered reverse direction the discovery effect-size baseline (0.57) in fact exceeds the ER anchor (0.56), so we cannot claim the anchor adds anything at all in this pair. We therefore do not claim the gate is uninformative in breast, only that its effect is small where glioma's is large, and undetectable against the appropriate baseline. Three non-exclusive reasons make breast diagnostic of *when the method helps*: the cohorts measure genes highly concordantly (genome-wide agreement of expression–ER correlations $r = 0.84$), leaving little artifact to flag; in ER-positive breast almost any gene set is prognostic [25], so non-replication reflects weak, proliferation-dominated signal and limited power (TCGA-BRCA has 67 events) rather than measurement fidelity; and the breast anchors (ER, basal subtype) differ from the glioma anchors (grade, *IDH*), confounding anchor informativeness with cohort concordance. One cohort pair cannot separate these, but all three predict the gate should add little—as observed, in contrast to glioma (§6.4, §6.8). The framework is therefore best understood not as a general replication oracle but as a **targeted detector of measurement-fidelity–driven discordance**: its predictive value scales with how much cross-cohort measurement heterogeneity exists, and it correctly adds little when both cohorts are clean.

7. Discussion

EGFR expression is a subtype artifact in TCGA: elevated in IDH-wildtype tumors, and once *IDH* or subtype is modeled it carries no independent prognostic information and adds no discrimination—robust to VST normalization and to an independent raw-read reprocessing (Tables S3, S5). Crucially the null extends *within* the IDH-wildtype subtype, where neither expression nor amplification predicts survival (Table 3). EGFR is therefore a between-subtype marker of the IDH-wildtype state—a more complete characterization than "confounded by IDH," since it rules out a residual within-subtype effect that grade-only adjustment could have missed.

Our attempt to validate in CGGA is instructive precisely because it initially appeared to contradict TCGA. Positive controls showed that CGGA EGFR expression tracks neither grade nor the canonical IDH relationship while CGGA clinical signals are strong, and the attenuation reproduced across three datasets and two platforms, with the CGGA-693 expression–clinical join verified sample by sample. We guarded against the obvious objection—that we discarded the cohort that disagreed—by specifying the criterion from biology in advance, testing the difference formally, and replicating across batches. A second objection, that CGGA simply contains a different patient mix, turns out to be partly right and is worth stating precisely: composition does explain a real share of the marginal grade discordance, and holding *IDH* fixed the EGFR–grade slopes agree. What it does not explain is the collapse in the fraction of EGFR's variance attributable to grade and *IDH*—from 0.106 to 0.024–0.045—with within-stratum variance unchanged (§6.4).

The methodological lesson is sharper, and more specific, than the one we set out to draw. The gate's apparent genome-scale performance (AUC 0.82–0.93) survives comparison with the baselines we chose but not with the ones we should have chosen: a same-population null reaches 0.89–0.93 with the same score, and discovery effect size alone reaches 0.74–0.78 (§6.9). The diagnosis is that the *product* form $r_D \times r_R$ rewards anchor magnitude, which tracks effect size, as much as cross-cohort agreement. The remedy is not to abandon the gate but to change the statistic: the difference form $-|r_D - r_R|$ has a null AUC of 0.53–0.61 rather than 0.79–0.90, and is positive in all three glioma pairs including the one where the product form is inverted. How strong that evidence is depends on how uncertainty is counted, and

Table 5: Matched-null calibration, on **Table 4's own gene universe** (four-cohort intersection, TCGA SD > 0.5, capped at the 8,000 most variable). Within each row the two arms share a discovery set, a gene universe, and a replication-cohort size by construction; "null" is a held-out, disjoint subsample of the discovery cohort, in which no cross-cohort artifact can exist. Intervals are paired bootstraps over *genes* (2,000 resamples), the sampling unit of an AUC; FDR is Benjamini–Hochberg within metric. A positive difference means the metric registers genuine cross-cohort discordance. The anchor *product* is heavily confounded (null 0.79–0.90) and changes sign across settings; the *difference* form is far less so (null 0.53–0.61) and its point estimate exceeds the null in all three glioma pairs, though once split variability is included only CGGA-325 remains conclusive (§6.9). The effect-size control, which cannot carry cross-cohort information, is negative throughout, as a negative control must be. Splitting the discovery cohort halves its size, so absolute AUCs are not interchangeable with Table 4's; the informative comparison is within a row. Restricting to the 2,000 most variable genes moves point estimates but changes no glioma conclusion except CGGA-693's difference form, which is positive but interval-spanning zero there (+0.051; −0.019 to +0.121). The intervals shown are within-split only: they resample genes from one split. A co-expression cluster bootstrap widens them by a median 2.2×, and combining six independent splits widens them a further 3.1–8.2×, after which only CGGA-325's difference form excludes zero (§6.9). Quote the combined intervals, not these.

Setting (discovery n / arm n / genes)	Metric	null	real	diff	95% CI	FDR
Glioma: TCGA→CGGA-693	product	0.900	0.824	−0.075	(−0.093, −0.058)	0.001
(368 / 367 / 5,495)	difference	0.530	0.669	+0.138	(+0.102, +0.176)	0.001
	effect size	0.820	0.785	−0.035	(−0.056, −0.013)	0.002
Glioma: TCGA→CGGA-325	product	0.893	0.937	+0.044	(+0.032, +0.055)	0.001
(513 / 222 / 6,037)	difference	0.611	0.796	+0.185	(+0.159, +0.210)	0.001
	effect size	0.844	0.767	−0.077	(−0.093, −0.060)	0.001
Glioma: TCGA→array-301	product	0.793	0.924	+0.132	(+0.120, +0.144)	0.001
(487 / 248 / 6,381)	difference	0.613	0.760	+0.148	(+0.129, +0.167)	0.001
	effect size	0.768	0.733	−0.036	(−0.051, −0.020)	0.001
Breast: METABRIC→BRCA	product	0.716	0.566	−0.150	(−0.207, −0.094)	0.001
(1,066 / 622 / 1,021)	difference	0.544	0.487	−0.057	(−0.118, +0.004)	0.062
	effect size	0.634	0.566	−0.068	(−0.125, −0.011)	0.019
Breast: TCGA-BRCA→METABRIC is *not evaluable* under this design at either universe: splitting 622 patients with 67 events leaves 26 prognostic genes at 2,000 genes and **none at all** at 8,000, too few for an AUC in either arm.						

counting it properly costs a great deal: split variability supplies 77–84% of the total variance, and once included only CGGA-325 excludes zero (§6.9). We were careful not to over-read this. The calibration problem is real but **not universal**: it bites hardest for the product form, which among the four evaluable cohort pairs is inverted in one and uninformative in the breast direction, while the difference form registers all three glioma pairs. One breast direction could not be evaluated at all—splitting 622 patients with 67 events leaves 26 prognostic genes at a 2,000-gene universe and none at 8,000. What is general is the requirement, not the verdict—split the discovery cohort, build a same-population replication set of matched size, and report the AUC against that rather than against 0.5, then judge each setting on what that comparison shows. This reframes cross-cohort validation as a two-part question—*does the association reproduce, and does the replication cohort measure the gene faithfully?*—while making clear that the second part needs a calibrated yardstick. The safeguard remains cheap and agnostic to gene, cohort, or platform, but *scoped* (§6.10): where both cohorts measure genes concordantly it has little to flag.

For cross-cohort prognostic studies we recommend: (1) identify a covariate known to associate with the gene (grade for oncogenes, hormone receptor status for breast cancer genes, subtype for pathway members); (2) verify it reproduces with consistent sign and magnitude in both cohorts before interpreting prognostic comparisons; and (3) flag discordant genes (correlations differing by Fisher r-to-z $p < 0.05$, opposite signs, or $|r| < 0.1$ in either cohort) for review or exclusion. The gate needs one known covariate and negligible computation.

8. Limitations

First, the primary finding rests on a single cohort: because the CGGA expression matrix could not measure EGFR faithfully, no successful external validation was obtained, so the conclusion is internally robust and biologically consistent but externally unreplicated for this gene. Second, the failure is reproducible (three datasets, two platforms) and not an error in our handling of the CGGA-693 RSEM matrix—join verified, result independently recomputed—but its cause within

Table 6: Controlled experiment with known ground truth: five mechanistically distinct measurement failure modes, injected into a random 20% of genes in one half of TCGA only (2,000 most-variable genes, 4 replicates per cell, no composition shift). AUC is scored against the genes actually corrupted. "Variance share" is the fraction of a corrupted gene's variance explained by the six (grade, *IDH*) strata afterwards; the uncorrupted baseline is 0.339. Compression—the signature EGFR shows in CGGA—is the best-detected mode; flooring is a blind spot. Replicate SDs of the AUC are 0.004–0.026 throughout.

Failure mode	Mechanism	Dose	Corrupt / clean replication	Variance share	AUC (detect corrupted)
Compression	shrink between-stratum signal by λ	$\lambda = 0.25$	0.43 / 0.93	0.046	**0.848**
Compression		$\lambda = 0.50$	0.80 / 0.95	0.136	**0.730**
Compression		$\lambda = 0.75$	0.91 / 0.96	0.236	0.639
Permutation	shuffle a fraction of values	25%	0.92 / 0.97	0.197	0.630
Permutation		50%	0.68 / 0.95	0.098	0.728
Permutation		75%	0.24 / 0.96	0.035	0.858
Additive noise	$x + N(0, k\,\mathrm{sd})$	$k = 0.5$	0.95 / 0.96	0.268	0.567
Additive noise		$k = 2.0$	0.54 / 0.93	0.079	0.779
Contamination	mix in an unrelated gene at weight w	$w = 0.25$	0.89 / 0.93	0.308	0.543
Contamination		$w = 0.75$	0.49 / 0.94	0.329	0.734
Flooring	censor below the qth quantile	$q = 0.25$	0.94 / 0.95	0.341	**0.498**
Flooring		$q = 0.75$	0.82 / 0.95	0.184	**0.599**

CGGA (normalization, gene model, or platform) was not identified. Third, adjusted models used complete cases without imputation, though the key nested comparison was refit on a common sample. Fourth, the pooled TCGA analysis mixes two separately processed studies, and grade IV is collinear with the glioblastoma study, so batch and grade cannot be separated. Fifth, the EGFR term violates PH in the *IDH*-adjusted lower-grade model, and the global PH test is violated in that model as well ($p = 3.6 \times 10^{-5}$), driven by grade and *IDH* rather than EGFR; period-specific estimates are null before three years and non-significantly elevated afterwards (§6.2), so a late-emerging effect cannot be excluded, though the EGFR term did satisfy PH in the within-subtype model ($p = 0.084$). Sixth, the systematic method was benchmarked against no-gating, detectability, and estimation-precision baselines but not a broad panel of alternative QC metrics, and "grade-discordant" flags a candidate—not proven—measurement artifact. Related and more serious, the genome-scale AUCs in Table 4 are not interpretable against chance: a same-population null cohort reaches 0.79–0.90 with the same product-form score on the same gene universe (§6.9), so those figures largely quantify how well the anchor ranks genes by signal strength. They are not, however, empty—on a single split the real cohort's point estimate exceeds its null in two of three glioma pairs, and the difference form's does so in all three—though once split variability is included only CGGA-325's difference form remains conclusive, and no glioma pair's product form significantly exceeds its null (§6.9). We report Table 4 for continuity with the screens in §6.7, but Table 5 is the interpretable version and the controlled experiment in Table 6 is what establishes that the gate detects mis-quantification. Tenth, that controlled experiment covers five failure modes but not batch effects or annotation errors, its doses are not calibrated to any real assay, and it establishes a blind spot for flooring/saturation that we have not characterized further. Eleventh, our own scripts compute the anchor correlation as Spearman in the screens and Pearson in the transport and baseline analyses, so anchor AUCs differ by up to 0.007 between them (0.815 vs. 0.808 for CGGA-693); comparisons are internally consistent within each analysis but should not be pooled across them. Twelfth, and the most consequential limitation of the calibration: uncertainty is dominated by which patients fall in which half, not by which genes are sampled. Combining $K = 6$ splits per setting attributes 77–84% of total variance to between-split variation and widens the intervals 3.1–8.2× relative to a single-split gene bootstrap; a co-expression cluster bootstrap adds a further median 2.2× on the within-split component. Under the combined accounting the difference form remains positive in all three glioma pairs and in all 18 split-level estimates, but only CGGA-325 excludes zero, array-301 is on the boundary, and CGGA-693 spans it (§6.9). The single-split intervals in Table 5 should therefore be read as within-split precision only. This is a limitation of cohort size as much as of method: with a few hundred patients per arm, a matched null cannot resolve differences of this magnitude, and only a materially larger discovery cohort would. The gene universe matters in one place: CGGA-693's difference-form result holds on the full universe but is interval-spanning on the 2,000 most variable genes. Thirteenth, the breast arm contributes one evaluable direction rather than two, so the claim that the calibration problem is setting-dependent rests on four cohort pairs, three of them from one consortium. Fourteenth, the matched-null calibration was run with the grade anchor only. It is fully robust to the discovery and replication thresholds (54 of 54 grid cells) but not to the anchor: with *IDH* substituted, 39 of 54 cells agree and CGGA-325's product form inverts throughout. Table 4's *IDH*-anchor column is therefore reported without a corresponding calibration, and should be read as uncalibrated. Seventh, the three replication datasets come from a single resource and are not fully independent (61 patients shared between array-301 and CGGA-325 and 6 with CGGA-693, quantified and controlled for in §6.8), and all replication is against one consortium in one tumor type. Eighth, and conceptually important, the framework assesses cross-cohort *reproducibility*, not *validity*: it can bless a reproducible association that is nonetheless confounded (as EGFR's is by *IDH*), so it complements rather than replaces causal scrutiny. Ninth, part of the genome-scale AUC reflects shared grade signal between the "prognostic" definition and the grade anchor; the effect persists but is weaker for subtype-independent associations (Table S7).

9. Future Work

True new-patient external validation still requires an independent cohort whose EGFR expression passes the positive controls. Reprocessing CGGA from raw reads would be the natural test, but its raw reads are controlled-access; an openly available, faithfully quantified glioma RNA-seq cohort with *IDH*, grade, and survival is therefore needed. For the framework, the informative next step is to evaluate cohort pairs that *do* differ in measurement fidelity across several tumor types—the regime the gate targets—and better-powered discovery cohorts, since the breast pair (§6.10) was both well-matched and underpowered. Because the observational AUC is now known to be uninformative about artifact detection, future evaluations of this or any competing gate should be reported against a split-half same-population null, and ideally against injected ground truth of the kind in Table 5, extended beyond permutation to saturation, batch, and annotation failure modes.

10. Conclusion

Cross-cohort prognostic replication in transcriptomics is unreliable, and a simple biological positive control does detect the subset of discordances that stem from faulty measurement—but establishing that required discarding the evidence we first assembled for it. Across 6,512 genes, three glioma cohorts, two platforms, and two anchors, a gene's expression–covariate concordance forecasts replication with AUC 0.82–0.93; a same-population null cohort, in which nothing can be wrong, forecasts it at 0.89–0.93 using the same score. That score is therefore dominated by signal strength, and the omitted effect-size baseline recovers much of it. The fix is a change of statistic rather than an abandonment of the method: scoring disagreement directly, as $-|r_D - r_R|$, drops the null to 0.53–0.61 and is positive in all three glioma pairs, though with cohorts of this size only one of the three is statistically conclusive once split variability is counted. What establishes that the gate detects real measurement failure is a controlled experiment: with corruption injected at known doses, it identifies compressed genes at AUC 0.73–0.85—compression being precisely EGFR's signature in CGGA—permuted and noised genes at 0.63–0.86, and floored genes not at all. Our recommendation is correspondingly specific: run the gate in its difference form, calibrate it against a split-half same-population null rather than against chance, and do not rely on it where saturation is the suspected failure mode. The framework is motivated by EGFR, prognostic in TCGA only until *IDH* is modeled, carrying no within-IDH-wildtype signal, and showing in all three CGGA datasets a 2–4× collapse in the share of its variance explained by grade and *IDH* that cohort composition does not account for. EGFR is therefore a marker of the IDH-wildtype state, not an independent prognostic factor.

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